

**Q1 - 24 January - Shift 2**

The equations of the sides AB and AC of a triangle ABC are

$$(\lambda + 1)x + \lambda y = 4 \text{ and } \lambda x + (1 - \lambda)y + \lambda = 0$$

respectively. Its vertex A is on the y-axis and its orthocentre is (1, 2). The length of the tangent from the point C to the part of the parabola  $y^2 = 6x$  in the first quadrant is

- (1)  $\sqrt{6}$
- (2)  $2\sqrt{2}$
- (3) 2
- (4) 4

Space for your notes:

**Q2 - 24 January - Shift 2**

The urns A, B and C contain 4 red, 6 black; 5 red, 5 black and  $\lambda$  red, 4 black balls respectively. One of the urns is selected at random and a ball is drawn. If the ball drawn is red and the probability that it is drawn from urn C is 0.4 then the square of the length of the side of the largest equilateral triangle, inscribed in the parabola  $y^2 = \lambda x$  with one vertex at the vertex of the parabola is

Space for your notes:

**Q3 - 25 January - Shift 1**

The distance of the point  $(6, -2\sqrt{2})$  from the common tangent  $y = mx + c$ ,  $m > 0$ , of the curves  $x = 2y^2$  and  $x = 1 + y^2$  is

- (1)  $\frac{1}{3}$
- (2) 5
- (3)  $\frac{14}{3}$
- (4)  $5\sqrt{3}$

Space for your notes:

**Q4 - 30 January - Shift 1**

If  $P(h, k)$  be point on the parabola  $x = 4y^2$ , which is nearest to the point  $Q(0, 33)$ , then the distance of  $P$  from the directrix of the parabola  $y^2 = 4(x + y)$  is equal to :

- (1) 2
- (2) 4
- (3) 8
- (4) 6

Space for your notes:

**Q5 - 30 January - Shift 2**

Let  $A$  be a point on the  $x$ -axis. Common tangents are drawn from  $A$  to the curves  $x^2 + y^2 = 8$  and  $y^2 = 16x$ . If one of these tangents touches the two curves at  $Q$  and  $R$ , then  $(QR)^2$  is equal to

- (1) 64
- (2) 76
- (3) 81
- (4) 72

Space for your notes:

**Q6 - 31 January - Shift 2**

Let  $S$  be the set of all  $a \in \mathbb{N}$  such that the area of the triangle formed by the tangent at the point  $P(b, c)$ ,  $b, c \in \mathbb{N}$ , on the parabola  $y^2 = 2ax$  and the lines  $x = b$ ,  $y = 0$  is  $16 \text{ unit}^2$ , then  $\sum_{a \in S} a$  is equal to \_\_\_\_\_.

Space for your notes:

**Q7 - 01 February - Shift 2**

If the x-intercept of a focal chord of the parabola  $y^2 = 8x + 4y + 4$  is 3, then the length of this chord is equal to \_\_\_\_\_.

Space for your notes:

## Answer Key

**(As per Official NTA Key released on 2 Feb)**

**#MathBoleTohMathonGo**

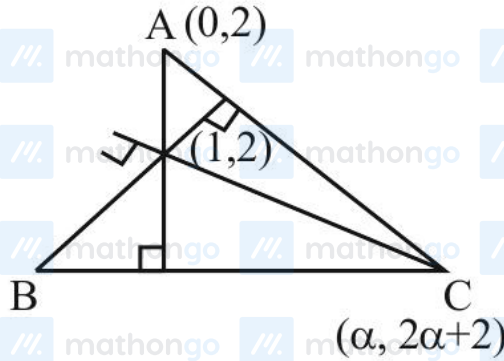
#MathBoleTohMathonGo

$$AB: (\lambda + 1)x + \lambda y = 4$$

$$AC: \lambda x + (1 - \lambda)y + \lambda = 0$$

Vertex A is on y-axis

$$\Rightarrow x = 0$$



$$\text{So } y = \frac{4}{\lambda}, y = \frac{\lambda}{\lambda - 1}$$

$$\Rightarrow \frac{4}{\lambda} = \frac{\lambda}{\lambda - 1}$$

$$\Rightarrow \lambda = 2$$

$$AB: 3x + 2y = 4$$

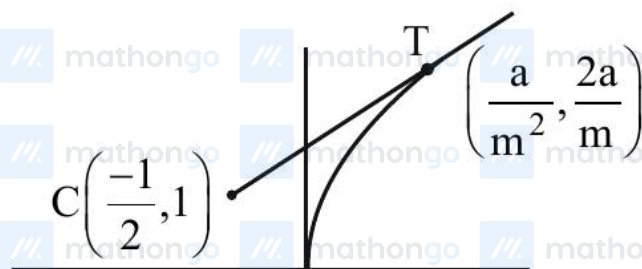
$$AC: 2x - y + 2 = 0$$

$$\Rightarrow A(0, 2) \text{ Let } C(\alpha, 2\alpha + 2)$$

$$\text{Now (Slope of Altitude through C)} \left( -\frac{3}{2} \right) = -1$$

$$\left( \frac{2\alpha}{\alpha - 1} \right) \left( -\frac{3}{2} \right) = -1 \Rightarrow \alpha = -\frac{1}{2}$$

$$\text{So } C\left(-\frac{1}{2}, 1\right)$$



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## Questions with Solutions

MathonGo

Let Equation of tangent be  $y = mx + \frac{3}{2m}$

Q2 (432)

$$m^2 + 2m - 3 = 0$$

$$\Rightarrow m = 1, -3$$

So tangent which touches in first quadrant at T is

$$T \equiv \left( \frac{a}{m^2}, \frac{2a}{m} \right)$$

$$\equiv \left( \frac{3}{2}, 3 \right)$$

$$\Rightarrow CT = \sqrt{4 + 4} = 2\sqrt{2}$$

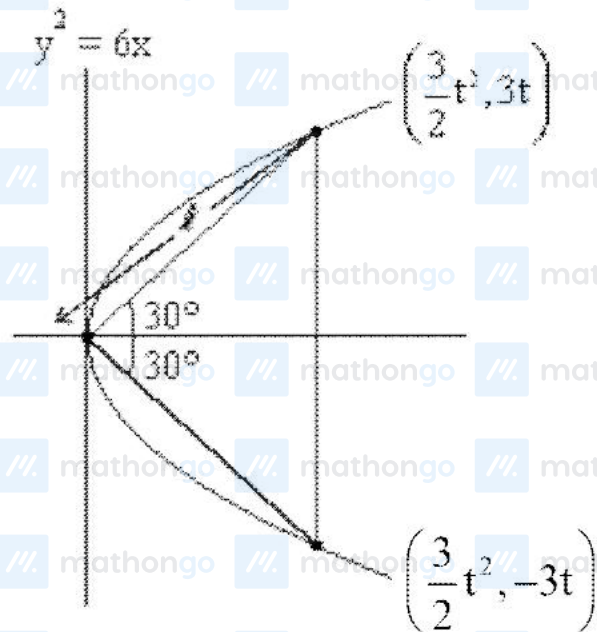
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| Urn A |       | Urn B |       | Urn C     |       |
|-------|-------|-------|-------|-----------|-------|
| Red   | Black | Red   | Black | Red       | Black |
| 4     | 6     | 5     | 5     | $\lambda$ | 4     |

$$P\left(\frac{C}{R}\right) = \frac{P(C)P\left(\frac{R}{C}\right)}{P(A)P\left(\frac{R}{A}\right) + P(B)P\left(\frac{R}{B}\right) + P(C)P\left(\frac{R}{C}\right)}$$

$$0.4 = \frac{\frac{1}{3} \times \frac{\lambda}{(\lambda+4)}}{\frac{1}{3} \times \frac{4}{10} + \frac{1}{3} \times \frac{5}{10} + \frac{1}{3} \times \frac{\lambda}{(\lambda+4)}}$$

$$\Rightarrow \lambda = 6$$



$$\tan 30^\circ = 3t = \frac{3}{2}t^2$$

$$\frac{1}{\sqrt{3}} = \frac{2}{t}$$

$$t = 2\sqrt{3}$$

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Q3 (2)

$$\rho^2 = 18^2 + (6\sqrt{3})^2$$

For

$$y^2 = \frac{x}{2}, T: y = mx + \frac{1}{8m}$$

For tangent to  $y^2 + 1 = x$

$$\Rightarrow \left( mx + \frac{1}{8m} \right)^2 + 1 = x$$

$$D = 0 \Rightarrow m = \frac{1}{2\sqrt{2}}$$

$$\therefore T: x - 2\sqrt{2}y + 1 = 0$$

$$d = \left| \frac{6 + 8 + 1}{\sqrt{9}} \right| = 5$$

Q4 (4)

Equation of normal

$$y = -tx + 2at + at^3$$

$$y = -tx + \frac{2}{16}t + \frac{1}{16}t^3$$

It passes through (0, 33)

$$33 = \frac{t}{8} + \frac{t^3}{16}$$

$$t^3 + 2t - 528 = 0$$

$$(t - 8)(t^2 + 8t + 66) = 0$$

$$t = 8$$

$$P(at^2, 2at) = \left( \frac{1}{16} \times 64, 2 \times \frac{1}{16} \times 8 \right) = (4, 1)$$

Parabola :

$$y^2 = 4(x + y)$$

$$\Rightarrow y^2 - 4y = 4x$$

$$\Rightarrow (y - 2)^2 = 4(x + 1)$$

Equation of directrix :-

$$x + 1 = -1$$

$$x = -2$$

Distance of point = 6

Ans. : (4)

Q5 (4)

$$y = mx + \frac{4}{m}$$

$$\frac{\left| \frac{4}{m} \right|}{\sqrt{1+m^2}} = 2\sqrt{2} \therefore m = \pm 1$$

$y = \pm x \pm 4$ . Point of contact on parabola

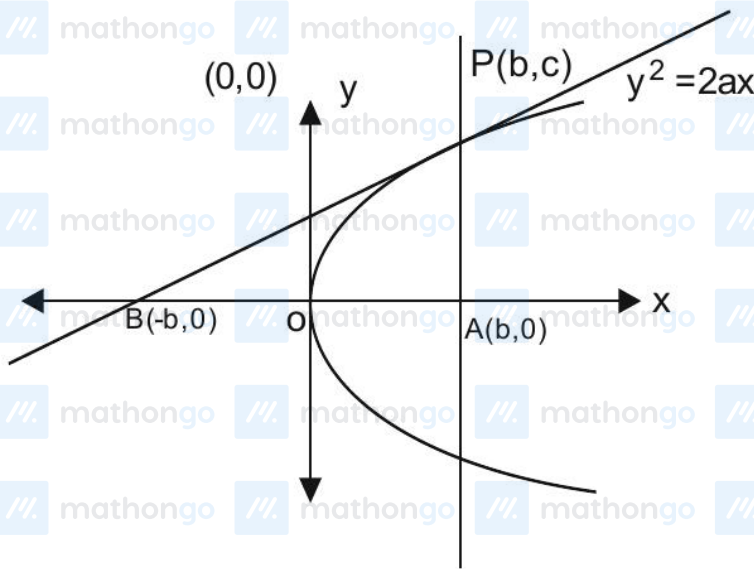
$$\text{Let } m = 1, \left( \frac{a}{m^2}, \frac{2a}{m} \right)$$

$$R(4, 8)$$

Point of contact on circle Q (-2, 2)

$$\therefore (QR)^2 = 36 + 36 = 72$$

Q6 (146)



As  $P(b, c)$  lies on parabola so  $c^2 = 2ab$  --- (1)

Now equation of tangent to parabola  $y^2 = 2ax$  in point

$$\text{form is } yy_1 = 2a \frac{(x + x_1)}{2}, (x_1, y_1) = (b, c)$$

$$\Rightarrow yc = a(x + b)$$

For point B, put  $y = 0$ , now  $x = -b$

So, area of  $\Delta PBA$ ,  $\frac{1}{2} \times AB \times AP = 16$

$$\Rightarrow \frac{1}{2} \times 2b \times c = 16$$

$$\Rightarrow bc = 16$$

As  $b$  and  $c$  are natural number so possible values of  $(b, c)$  are  $(1, 16)$ ,  $(2, 8)$ ,  $(4, 4)$ ,  $(8, 2)$  and  $(16, 1)$

Now from equation (1)  $a = \frac{c^2}{2b}$  and  $a \in \mathbb{N}$ , so

values of  $(b, c)$  are  $(1, 16)$ ,  $(2, 8)$  and  $(4, 4)$  now

values of are 128, 16 and 2.

Hence sum of values of  $a$  is 146.

Q7 (16)

$$y^2 = 8x + 4y + 4$$

$$(y - 2)^2 = 8(x + 1)$$

$$y^2 = 4ax$$

$$a = 2, X = x + 1, Y = y - 2$$

focus (1, 2)

$$y - 2 = m(x - 1)$$

Put (3, 0) in the above line

$$m = -1$$

Length of focal chord = 16